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$$\lim_{x\to 0} \frac{x^2 \ln(1+x)}{\ln(1-x)} = \frac{0}{0}$$

$$\overset{H}{=} \lim_{x\to 0} \frac{2x\ln(1+x) + x^2 \cdot \dfrac{1}{1+x}}{\cos x - 1}$$

$$= \lim_{x\to 0} \frac{2x\ln(1+x) + \dfrac{x^2}{1+x}}{\cos x - 1}$$

$$\overset{H}{=} \lim_{x\to 0} \frac{2\ln(1+x) + \dfrac{2x + 2x^2 + 2x + 2x^2 - x^2}{(1+x)^2}}{-\sin x}$$

$$= \lim_{x\to 0} \frac{2\ln(1+x) + \dfrac{4x + 3x^2}{(1+x)^2}}{-\sin x}$$

$$\overset{H}{=} \lim_{x\to 0} \frac{\dfrac{2}{1+x} + \dfrac{(4+6x)(1+2x) - 2(4x+3x^2)}{(1+x)^3}}{-\cos x}$$

$$= \lim_{x\to 0} \frac{\dfrac{2}{1+x} + \dfrac{2x+4}{(1+x)^3}}{-\cos x}$$

$$= \lim_{x\to 0} \frac{\dfrac{2x^2 + 6x + 6}{(1+x)^3}}{-\cos x} = \frac{6}{-1} = -6$$

References